Chapter 5 Notes

Section1

Notes

After Corollary 1.9 it reads: "We thus see that the function σ is strictly progressing." Certainly, this was shown for *ordinals*, but we have not shown this for any sets in general. In fact, σ being strictly progressing for all sets is equivalent to $x \in x$ not holding for any set x. For, on the one hand, if σ is strictly progressing for all sets, suppose $x \in x$ for some set x. Now, $x^+ = x \cup \{x\}$. Also, $x \subset x^+$ (by assumption). But consider any element y of x^+. Either $y \in x$ or $y = x$. In either case, because we assumed $x \in x$, $y \in x$. Thus $x^+ \subseteq x$, contrary to the assumption that $x \subset x^+$. Hence $x \neq x$ for all sets x. On the other hand, suppose that $x \neq x$ for all sets x. Then it must be that $x \subset x^+$. For obviously $x \subseteq x^+$, and if $x = x^+$, then, because $x \in x^+$, $x \in x$, a contradiction. Thus σ would be strictly progressing.

Hence σ being strictly progressing for all sets is equivalent to $x \in x$ holding for no sets, which we have not shown.

In the proof of Theorem 1.17, it is said that no two distinct elements (of On) can have the same successor. To prove this, we first prove the lemma that (for any two ordinals), if $\beta < \alpha$, then $\beta^+ \neq \alpha^+$. For suppose $\beta < \alpha$ and $\beta^+ = \alpha^+$. Thus $\alpha \cup \{\alpha\} = \beta \cup \{\beta\}$. Thus either $\alpha = \beta$ or $\alpha \in \beta$. The first possibility is ruled out by the assumption that $\beta < \alpha$. Also, if $\alpha \in \beta$, the, by Theorem 1.12, $\alpha < \beta$, contrary to $\beta < \alpha$. Thus $\alpha \notin \beta^+$, a contradiction. Hence $\beta^+ \neq \alpha^+$.

Now suppose $\alpha^+ = \beta^+$. Thus $\alpha \cup \{\alpha\} = \beta \cup \{\beta\}$. Thus, if $x \in \alpha$, then either $x \in \beta$ or $x = \beta$. If $x = \beta$, then, because α and β are ordinals (and by Theorem 1.12), $\beta < \alpha$. But, by the above lemma, this means that $\beta^+ \neq \alpha^+$, a contradiction. Hence $x \neq \beta$, meaning $x \in \beta$ (and thus $\alpha \subseteq \beta$). Similar reasoning shows that $\beta \subseteq \alpha$, meaning $\alpha = \beta$, therefore completing the proof.

In Theorem 1.19, it was remakred that $\alpha < \beta$ implies that $\alpha^+ \leq \beta$. For suppose that $\alpha < \beta$. Then, by Theorem 1.12, $\alpha \in \beta$. Hence $\alpha \in \beta$ and for all $x \in \alpha$, $x \in \beta$, meaning $\alpha^+ \leq \beta$.

In Theorem 1.20, it is remarked that $\beta < \alpha^+$ implies that $\beta \leq \alpha$. Here is why that is: For suppose that $\beta < \alpha^+$. Thus every element of β is either an element of α, or is equal to α. Suppose some element x of β was equal to α. Because every ordinal is transitive, that would mean, for every $y \in \alpha$, $y \in \beta$. Hence $\beta = \alpha \cup \{\alpha\}$, contrary to the fact that $\beta < \alpha^+$. Thus all elements of β are elements of α (and none are equal to α) meaning $\beta \leq \alpha$.

Section 2

Notes

Note that Theorem 2.3 presupposes that x is a set, as this means that there is at least one ordinal outside of x (because On is not a set).

Exercise 2.1

This is an easy consequence of Theorem 2.2. For, by Theorem 2.2, every transitive subset of On is an element of On, and thus On is T-closed. Hence, if x belongs to all T-closed classes, then $x \in On$, meaning x is an ordinal.

Exercise 2.2

This is another easy consequence of Theorem 2.2. For consider any non-empty set A of ordinals. Let $B = \cap A$. Obviously, B is a set of ordinals. It suffices to show that B is transitive (and hence is an ordinal by Theorem 2.2). Consider any $x \in B$. Thus, for every $y \in A$, $x \in y$. But, because y is an ordinal (and thus it is transitive), all elements z of x are in y. Thus all elements z of x are in all elements of A, meaning all elements z of x are in $\cap A$. Thus B is transitive, meaning it is an ordinal.

Exercise 2.3

(a): On the one hand, we show that every ordinal x (which is of course a set) can be well ordered in such a way that each element y of x is the set of all elements of x that precede y in the ordering. We show that every ordinal can be well ordered under $\subseteq$ so that the required conditions are satisfied. We prove this using superinduction. First, recall that On is well ordered under $\subseteq$.
(1): Clearly $\emptyset$ is well ordered under $\subseteq$ such that the required condition is satisfied (in fact, $\subseteq\restriction \emptyset = \emptyset$).
(2): Now suppose α is well ordered such that the required condition is satisfied. Now consider α^+. It has already been shown that α^+ is a transitive set of ordinals. Thus, because it is a subclass of On, it is well ordered under $\subseteq$ (because On is well ordered under $\subseteq$). We need to show that $\subseteq\restriction \alpha^+$ satisfies the required condition. If $x \in \alpha^+$, either $x \in \alpha$ or $x = \alpha$ (because $\alpha^+ = \alpha \cup \{\alpha\}$). If $x \in \alpha$, then x is the set of all lesser elements in α by assumption. But it has already been shown that $x < \alpha$ (because $x \in \alpha$ iff $x < \alpha$), meaning x is the set of all elements of α^+ less than x. If $x = \alpha$, then obviously x is the set of all lesser elements of α^+. Hence, for every $x \in \alpha^+$, x is the set of all lesser elements of α^+.
(3): Lastly, suppose that some chain C of ordinals is well ordered under $\subseteq$ and satisfies the required conditions. Thus, by assumption, every element $x \in C$ is the set of all lesser elements of C. Consider any element $y \in \cup C$. Then it is an element of some $x \in C$, and thus, by assumption, it is also an element of C. Hence, by assumption, y is the set of all lesser elements of C. Thus every element in $\cup C$ is an element of C, meaning $\cup C \subseteq C$ and thus $\cup C$ is well ordered under $\subseteq$ (as C was assumed to be well ordered under $\subseteq$). Now we show that for any $y \in \cup C$ (and recall that we have also shown that $y \in C$), the set of all elements in $\cup C$ less than y is equal to the set of all elements in C less than y. For suppose x is an element in $\cup C$ less than y. Then, as we have shown, x is also an element in C. Thus $x \in C$, meaning x is a lesser element of y that is in C. On the other hand, if x is an element in C less than y, then, because $y \in C$, $x \in y$, meaning $x \in \cup C$. Thus we have shown that, for every $y \in \cup C$, it is the set of all lesser elements in C, and also the set of all lesser elements in $\cup C$. Thus $\subseteq\restriction \cup C$ satisfies the required condition.

Hence, by superinduction, it follows that every ordinal is well ordered and satisfies the given condition under $\subseteq$.

(b): On the other hand, we show that every set x that can be well ordered under $\leq$ in such a way that each element y of x is the set of all elements of x that precede y in the ordering, is an ordinal. It can be shown by transfinite induction on x that every element y of x is an ordinal. For suppose that for all elements z less than y, z is an ordinal. Then, by definition of x, y is the set of all lesser elements of x. Thus y is a set of ordinals. And y is transitive (because if $z \in y$ and $v \in z$, then $z \in x$, and thus $v < z$ and $z < y$, meaning $v < y$ (because $\leq$ is transitive) and thus $v \in y$). Thus y is a transitive set of ordinals, meaning it is an ordinal. Thus every element of x is an ordinal. Obviously x is transitive (because, by definition of x, if $y \in x$ and $z \in y$, then $z < y$ and $z \in x$). Thus x is a transitive set of ordinals, meaning it is an ordinal. This completes the proof.

Section 3

Notes

It is remarked that ω is a limit ordinal. Here is why that is. Obviously it is not the least ordinal. Suppose that it was a successor ordinal. Then, for some α, $\alpha < \omega$, and for no β is it the case that $\alpha < \beta < \omega$. Hence, by Theorem 1.12, $\alpha \in \omega$. But then $\alpha = n$ for some natural number n. Thus n^+ is another ordinal. But also (by Corollary 1.9) $n < n^+$ (and obviously $n^+ < \omega$, because all elements of n^+ are themselves natural numbers). Thus $\alpha < n^+ < \omega$, a contradiction to the fact that ω is the successor of α. Hence ω is not a successor ordinal nor the least ordinal, and consequently, ω is a limit ordinal.

It is remakred that $\omega \times 2$ is a transitive class of ordinals. Here is why this is. For $n > 1$,

$$\omega \times n = 0, 1, 2, ..., \omega, \omega + 1, \omega + 2, ..., \omega \times 2, \omega \times 2 + 1, \omega \times 2 + 2, ..., \omega \times (n-1), \omega \times (n-1) + 1, \omega \times (n-1) + 2, ...$$

and $\omega \times 1 = \omega$. From this definition it is obvious that $\omega \times n \subset \omega \times m$ if $m > n$. Also, we can show that if $n > m$, then $\omega \times a + m \subset \omega \times a + n$. This can be shown by induction on n. We let $\omega \times x + 0 = \omega \times x$. Then the base case vacuously holds (as there are no $m < 0$). Now suppose the result holds for some n. Then

$$\omega \times a + (n+1) = \omega \times a \cup \{\omega \times a + n\}$$

So suppose $m < n + 1$. Then either $m < n$ or $m = n$. In the first case, by assumption, because $\omega \times a + n \subset \omega \times a + (n+1)$ (because if this was not the case, this would imply that $\omega \times a + n \in \omega \times a + n$, contradicting the fact that $\omega \times a + n$ is an ordinal), and $\omega \times a + m \subset \omega \times a + n$, it follows that $\omega \times a + m \subset \omega \times a + (n+1)$. Then, if $m = n$, because $\omega \times a + n \subset \omega \times a + (n+1)$, $\omega \times a + m \subset \omega \times a + (n+1)$. Hence, in any case, $\omega \times a + m \subset \omega \times a + (n+1)$.

An obvious induction argument shows that, if $n < m$, then $\omega \times n + x \subset \omega \times m$ for all x. Then we have, if $m < n$, then $\omega \times m + x \subset \omega \times n + y$ for all x and y. From this it immediately follows that $\omega \times x + y$ is transitive for every x and y. For, by definition, if $a \in \omega \times x + y$ and $b \in a$, then either a is a natural number of $a = \omega \times m + n$ for some $m < x$ and n. If b is an element of a natural number, then it is also a natural number, so that $b \in \omega \times x + y$. If $b \in \omega \times m + n$, then, because $\omega \times m + n \subset \omega \times x + y$, $b \in \omega \times x + y$.